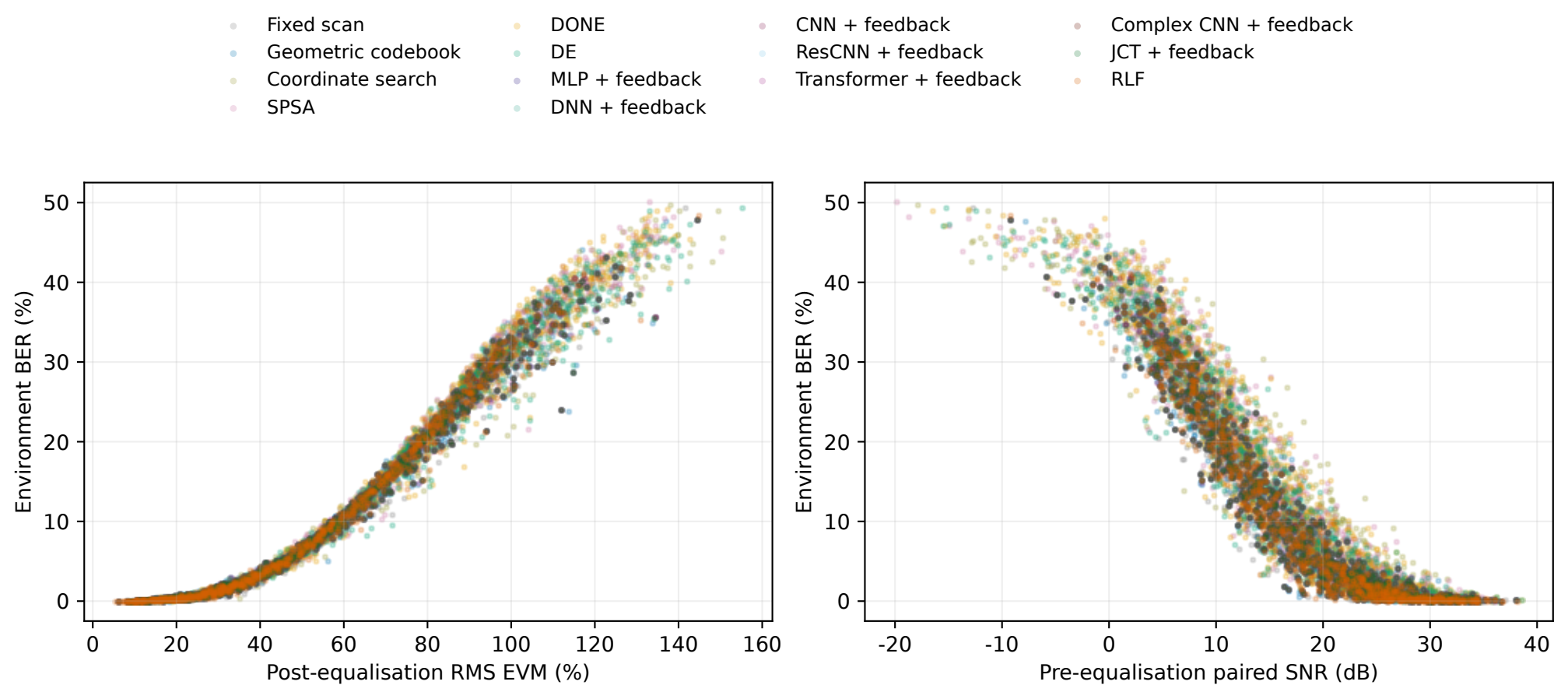
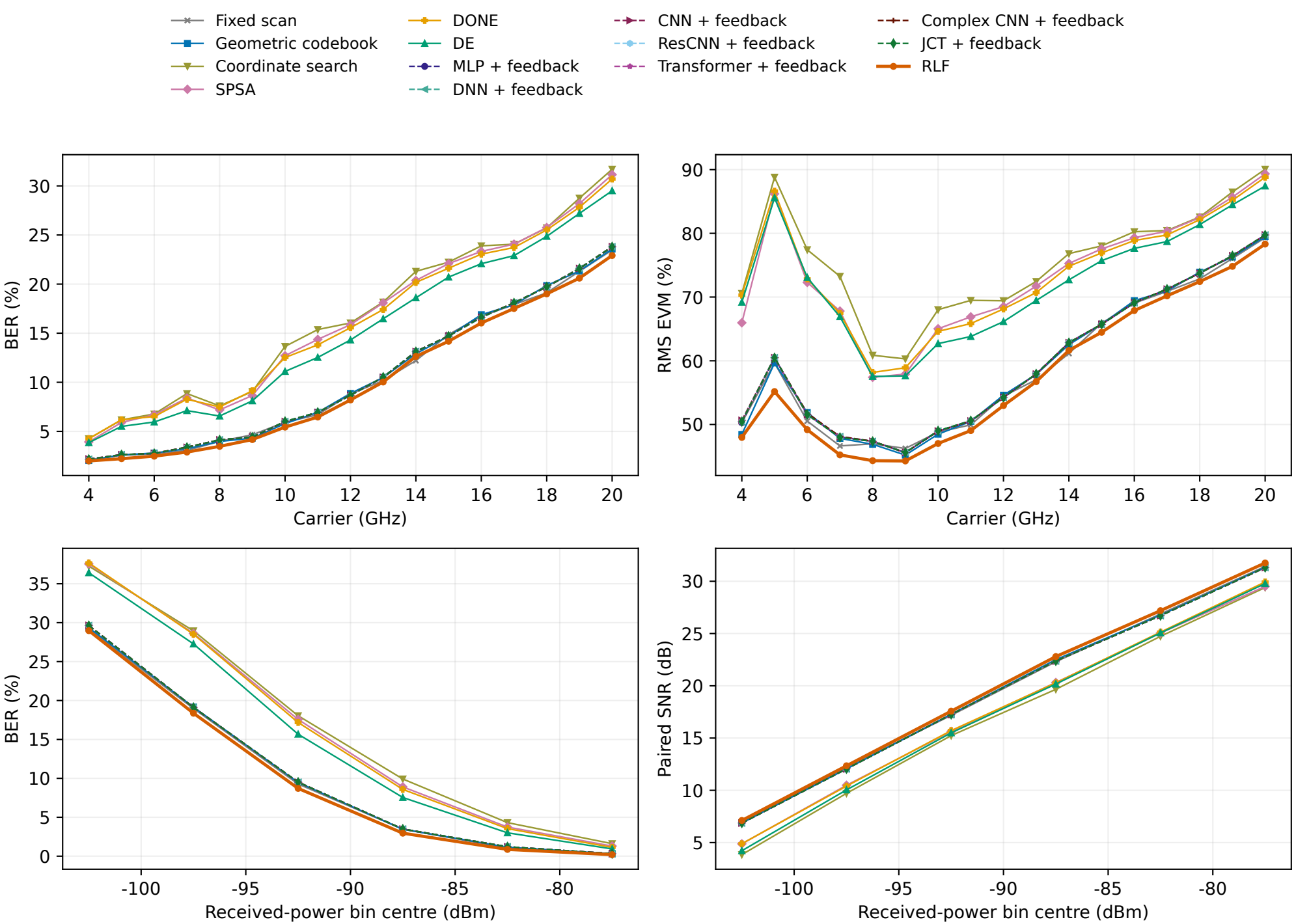
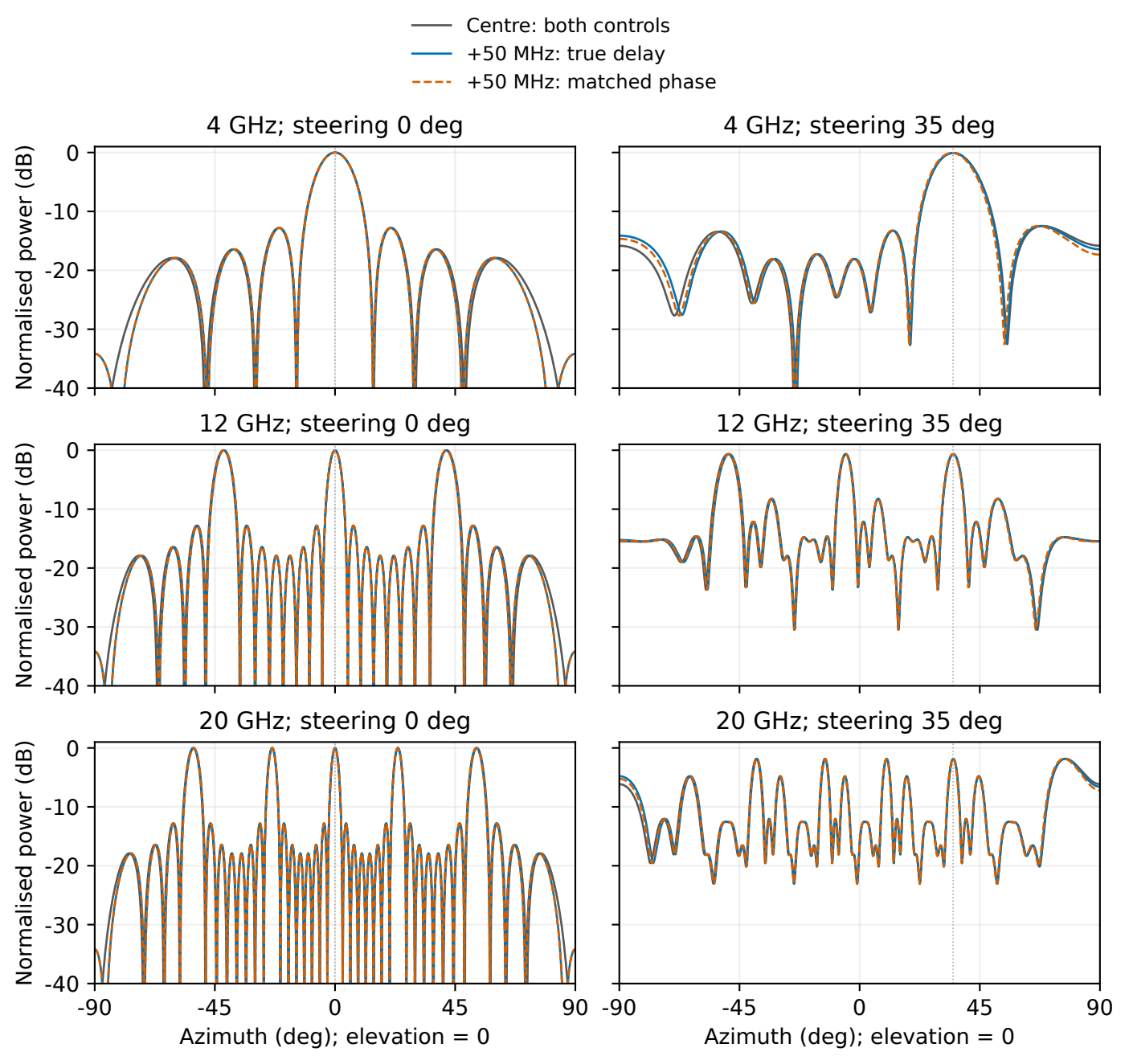
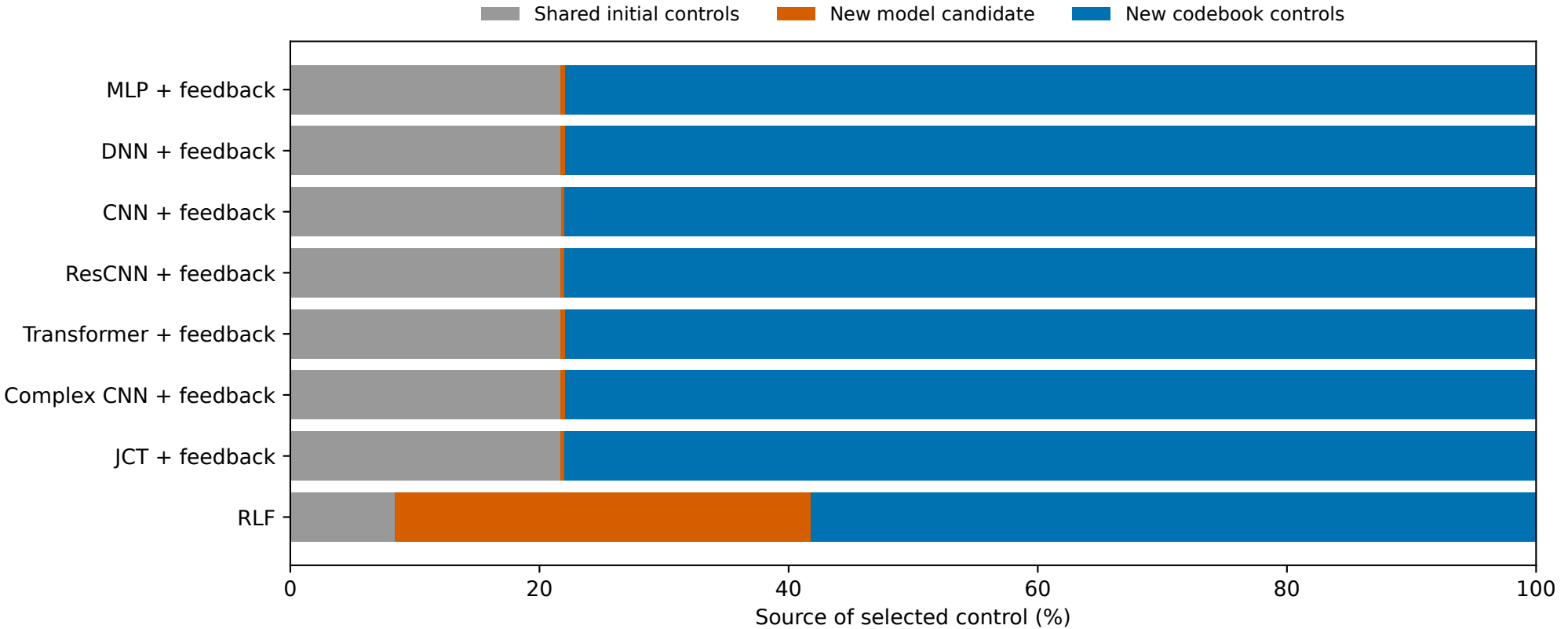
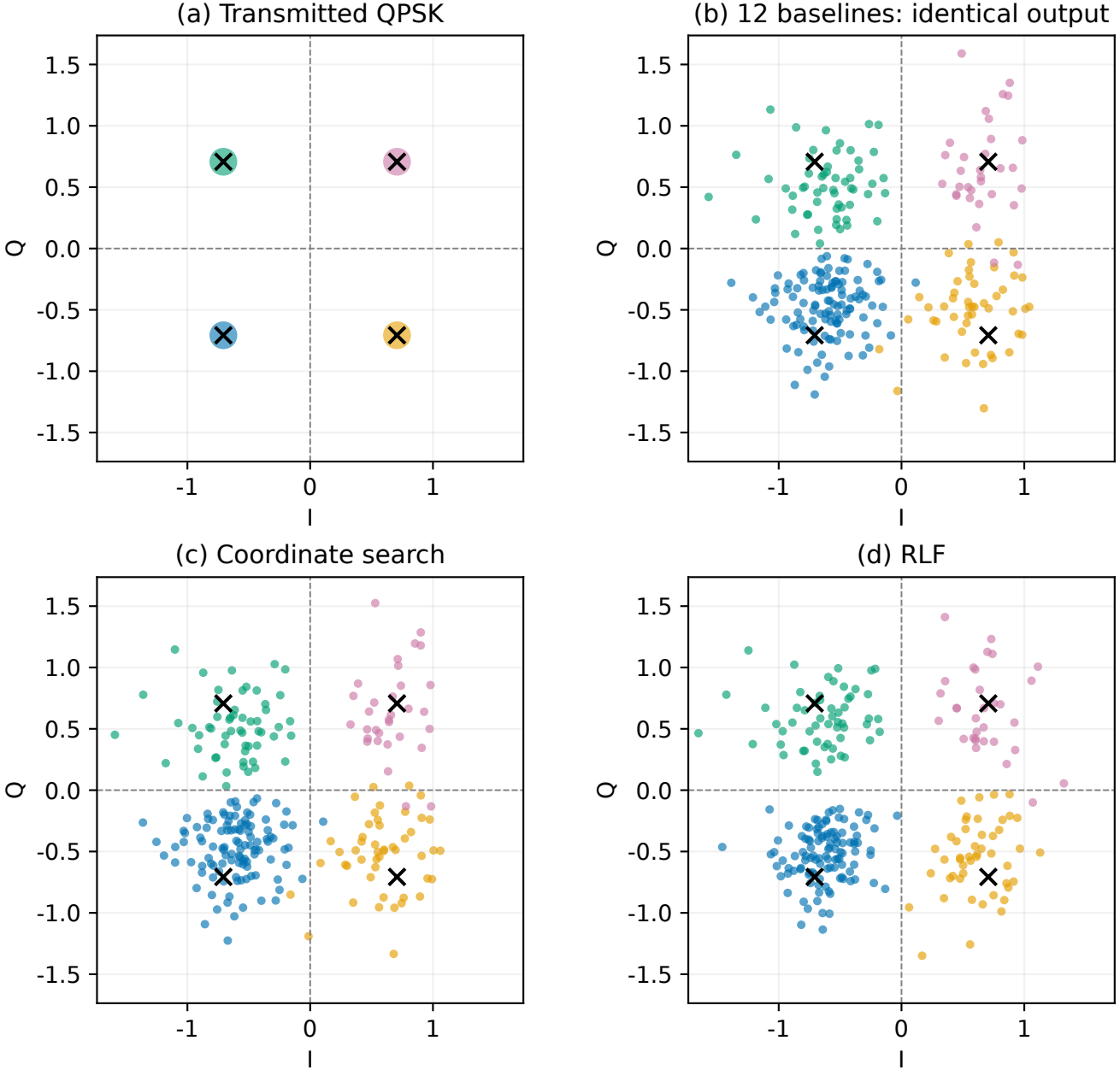
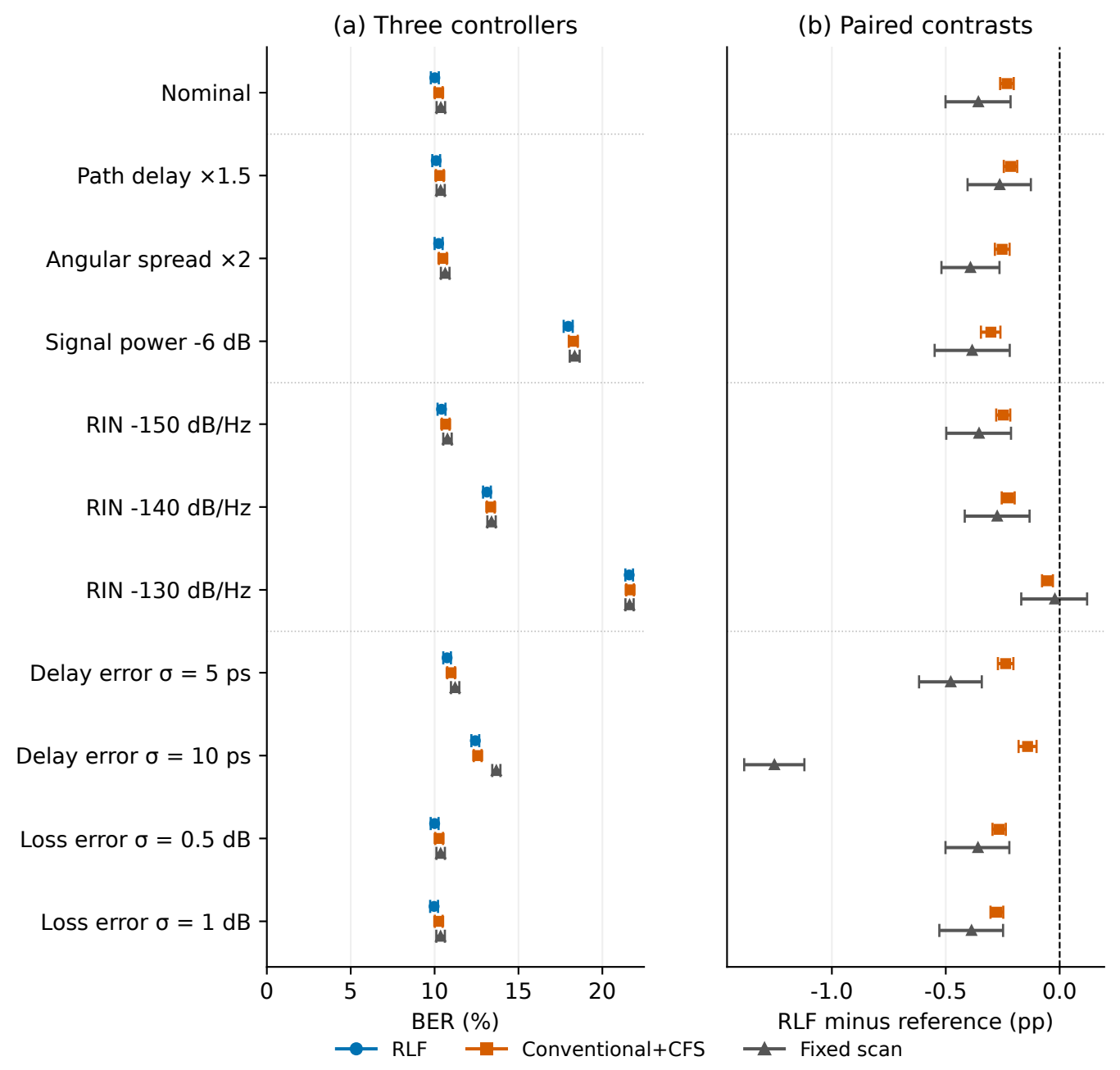

Fixed scan
Geometric codebook
Coordinate search
SPSA
DONE
DE
MLP + feedback
DNN + feedback
CNN + feedback
ResCNN + feedback
Transformer + feedback
Complex CNN + feedback
JCT + feedback
RLF
Environment BER (%)
Post-equalisation RMS EVM (%)
Environment BER (%)
Pre-equalisation paired SNR (dB)


Fixed scan
Geometric codebook
Coordinate search
SPSA
DONE
DE
MLP + feedback
DNN + feedback
CNN + feedback
ResCNN + feedback
Transformer + feedback
Complex CNN + feedback
JCT + feedback
RLF
BER (%)
Carrier (GHz)
RMS EVM (%)
Carrier (GHz)
BER (%)
Received-power bin centre (dBm)
Paired SNR (dB)
Received-power bin centre (dBm)

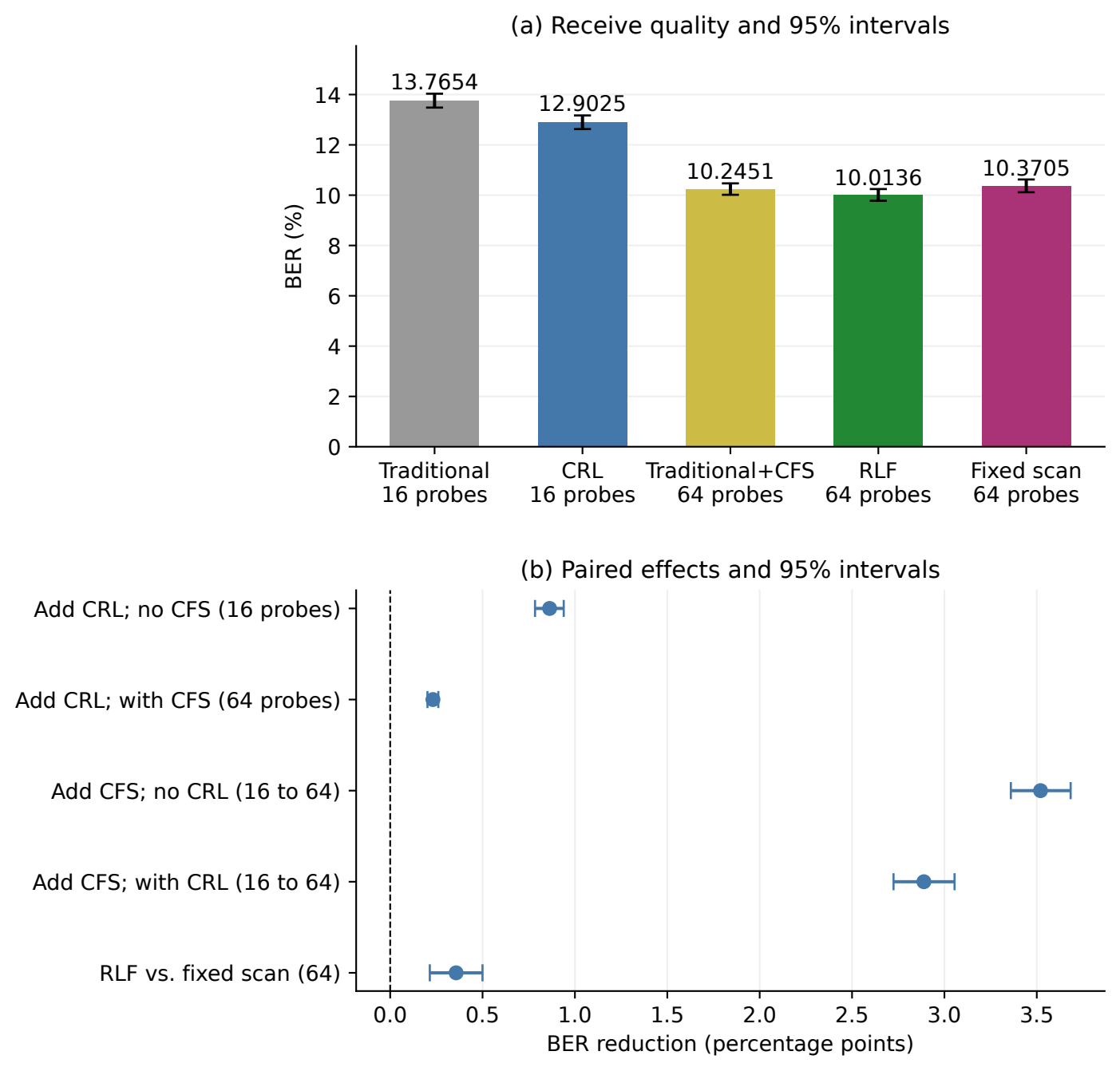

Centre: both controls
+50 MHz: true delay
+50 MHz: matched phase
4 GHz; steering 0 deg
4 GHz; steering 35 deg
12 GHz; steering 0 deg
12 GHz; steering 35 deg
20 GHz; steering 0 deg
20 GHz; steering 35 deg
Normalised power (dB)
0
-10
-20
-30
-40
-90
-45
0
45
90
Azimuth (deg); elevation = 0

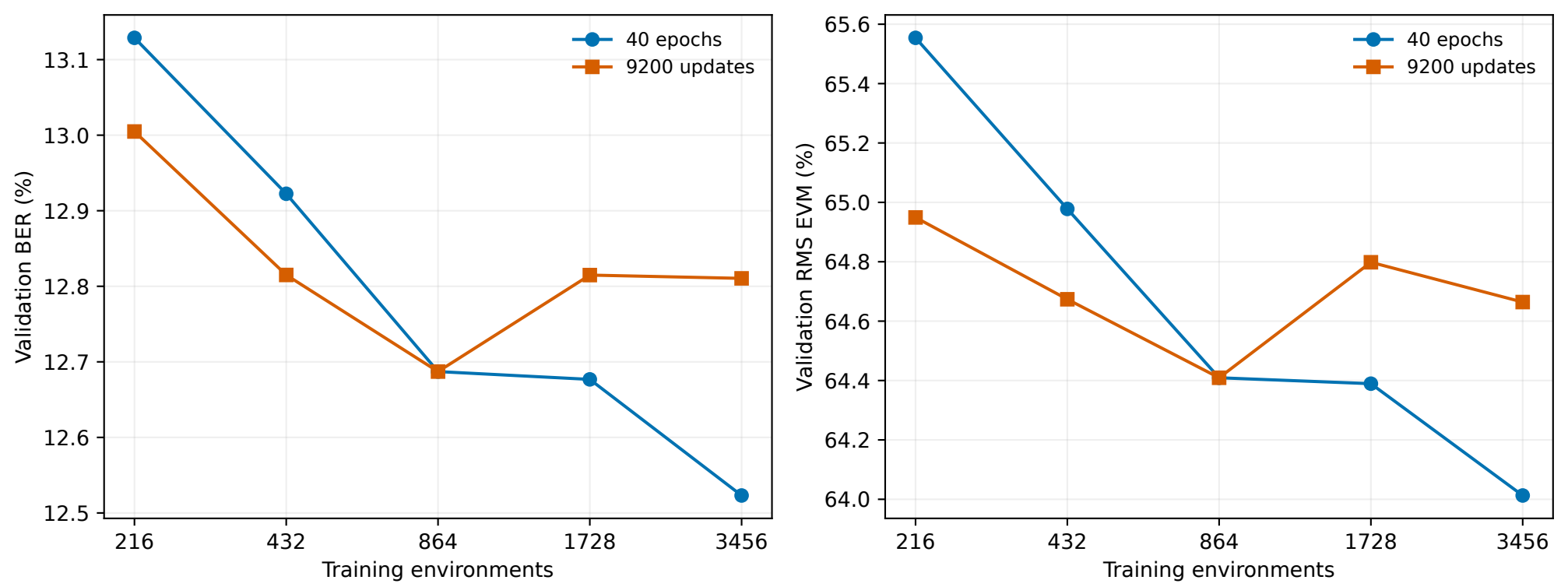

Shared initial controls
New model candidate
New codebook controls
MLP + feedback
DNN + feedback
CNN + feedback
ResCNN + feedback
Transformer + feedback
Complex CNN + feedback
JCT + feedback
RLF
0
20
40
60
80
100
Source of selected control (%)


(a) Transmitted QPSK
(b) 12 baselines: identical output
(c) Coordinate search
(d) RLF
I
Q
1.5
1.0
0.5
0.0
-0.5
-1.0
-1.5
-1
0
1



64 probes; paired 95% environment intervals.


(a) Receive quality and 95% intervals
BER (%)
0
2
4
6
8
10
12
14
13.7654
12.9025
10.2451
10.0136
10.3705
Traditional
16 probes
CRL
16 probes
Traditional+CFS
64 probes
RLF
64 probes
Fixed scan
64 probes
(b) Paired effects and 95% intervals
Add CRL; no CFS (16 probes)
Add CRL; with CFS (64 probes)
Add CFS; no CRL (16 to 64)
Add CFS; with CRL (16 to 64)
RLF vs. fixed scan (64)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
BER reduction (percentage points)


40 epochs
9200 updates
Validation BER (%)
13.1
13.0
12.9
12.8
12.7
12.6
12.5
216
432
864
1728
3456
Training environments
40 epochs
9200 updates
Validation RMS EVM (%)
65.6
65.4
65.2
65.0
64.8
64.6
64.4
64.2
64.0
216
432
864
1728
3456
Training environments